

# Learn HTML: Beginner's Guide

## Lesson 4: Creating Tables

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### Creating Tables - Study Notes

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#### Key Concepts

1. **Table Structure Elements** – `<table>`, `<tr>`, `<th>`, `<td>` define the skeleton of an HTML table.
2. **Rows vs. Cells** – `<tr>` creates a row; `<th>` (header cell) and `<td>` (data cell) populate that row.
3. **Adding Borders** – Simple borders can be applied with the `border` attribute (HTML) or, preferably, with CSS.
4. **Header vs. Body** – `<thead>`, `<tbody>`, and `<tfoot>` (optional) help organize large tables and improve accessibility.
5. **Accessibility Basics** – Use `<caption>` for a table title and scope attributes on `<th>` for screen reader clarity.

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#### Summary

HTML tables are built by nesting a series of tags that describe rows and cells. The outermost `<table>` element encloses the entire grid. Inside it, each `<tr>` tag creates a new horizontal line (row). Within a row, `<th>` tags define header cells—typically bold and centered by default—while `<td>` tags hold regular data.

To give a table a visible outline without complex styling, you can add a `border="1"` attribute directly to the `<table>` tag, though modern practice favors CSS (`border: 1px solid #000;`). Adding a `<caption>` provides a concise description, and using `<thead>`/`<tbody>` sections improves semantics and makes it easier to style or manipulate parts of the table with CSS or JavaScript. Simple tables are ideal for displaying small data sets, schedules, or comparison charts in beginner level web pages.

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#### Important Points

- **Basic Syntax**

```
```html
<table>
  <tr>          <!-- Row -->
    <th>Header</th> <!-- Header cell -->
    <td>Data</td>  <!-- Data cell -->
  </tr>
</table>
```
```

- **Border Options**

- HTML attribute: `<table border="1">` (quick demo, not recommended for production).

- CSS: `style="border: 1px solid #000; border-collapse: collapse;"` or an external stylesheet.
  - `border-collapse` removes double borders between cells, giving a cleaner look.
  - **Header Scope** – Add `scope="col"` or `scope="row"` to `<th>` for better accessibility.
  - **Caption** – `<caption>Monthly Sales Report</caption>` appears above the table by default.
  - **Semantic Grouping** – `<thead>`, `<tbody>`, `<tfoot>` are optional but useful for larger tables.

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## Examples

### Example 1 – Simple 2×3 Table with HTML Border Attribute

```
```html
<table border="1">
  <caption>Team Scores</caption>
  <tr>
    <th>Player</th>
    <th>Games</th>
    <th>Points</th>
  </tr>
  <tr>
    <td>Alice</td>
    <td>5</td>
    <td>42</td>
  </tr>
  <tr>
    <td>Bob</td>
    <td>5</td>
    <td>37</td>
  </tr>
</table>
```
```

\*Explanation:\*

- `border="1"` draws a thin line around each cell.
- The first `<tr>` uses `<th>` for column headings.
- Subsequent rows use `<td>` for regular data.

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### Example 2 – CSS Styled Table with Header Scope and Caption

```
```html
<style>
table {
  width: 60%;
  border: 2px solid #333;
  border-collapse: collapse;
}
```

```

    margin: 1rem 0;
}
th, td {
    border: 1px solid #666;
    padding: 0.5rem;
    text-align: left;
}
th {
    background-color: #f2f2f2;
}
</style>
<table>
  <caption>Quarterly Revenue (USD)</caption>
  <thead>
    <tr>
      <th scope="col">Quarter</th>
      <th scope="col">2023</th>
      <th scope="col">2024</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <th scope="row">Q1</th>
      <td>12,500</td>
      <td>13,200</td>
    </tr>
    <tr>
      <th scope="row">Q2</th>
      <td>14,300</td>
      <td>15,100</td>
    </tr>
  </tbody>
</table>

```

\*Explanation:\*

- CSS handles all border and spacing details, giving a cleaner look.
- `border-collapse: collapse;` merges adjacent borders.
- `scope="col"` and `scope="row"` clarify which cells are headers for assistive technologies.
- `<thead>` and `<tbody>` separate header rows from data rows, making future styling or scripting easier.

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## Quick Review

1. **What tag creates a new row inside a table?**
2. **Which element is used for header cells, and why might you add a `scope` attribute?**
3. **How do you apply a simple border to a table using CSS?**
4. **Name two optional tags that help organize larger tables.**
5. **What is the purpose of the `<caption>` element?**

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## Further Reading

- **CSS Table Styling:** `border-spacing`, `border-collapse`, alternating row colors (`nth-child`).
- **Responsive Tables:** Using `overflow-x:auto;` and CSS Grid/Flexbox alternatives.
- **Accessibility:** ARIA roles for tables, `summary` attribute (HTML5), and best practices for screen readers.
- **Advanced Table Features:** `<colgroup>` and `<col>` for column specific styling, merging cells with `rowspan`/`colspan`.

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\*End of notes – happy coding!\*

